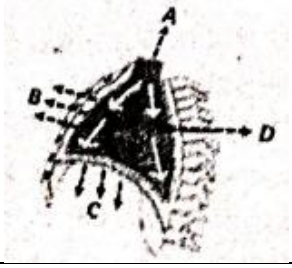
Question	The network of air tubes in insects for gaseous exchange is called as
Type	multiple choice
Option	Tracheal tubes
Option	Spiracles
Option	Stomata
Option	Skin
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Among vertebrates, fishes use ____ whereas amphibians, reptiles, birds and mammals respire through _____
Type	multiple choice
Option	Skin, Lungs
Option	Gills, Cuticle
Option	Gills, Lungs
Option	Gills, Spiracle
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The harmful gas produced by catabolic reaction (respiration) in our body is
Type	multiple choice
Option	Oxygen
Option	Carbon dioxide
Option	Phosphine
Option	Ozone
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Amphibians e.g. frogs respire:
Type	multiple choice
Option	Through moist skin
Option	Lungs
Option	Both a and b
Option	Trachea
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following organisms use vascularized bags for the exchange of gases?
Type	multiple choice
Option	Terrestrial organisms
Option	Aquatic organisms
Option	Both (a) & (b)
Option	None of the above
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The respiratory organs vary in different animals due to their

Type	multiple choice
Option	Nutrition
Option	Habitat and level of organization
Option	Excretion
Option	Reproduction
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Assertion - Amphibians can respire through moist skin also Reason - Frogs can shows both pulmonary and cutaneous respiration in water.
Type	multiple choice
Option	If both the assertion and the reason arc true and the reason is a correct explanation of the assertion.
Option	If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
Option	If the assertion is true but the reason is false.
Option	If both the assertion and reason are false.
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Respiratory organs in aquatic arthropods like crayfish, prawns, and molluscs like Unio are
Type	multiple choice
Option	Body wall
Option	Lungs
Option	Trachea
Option	Gills
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which group of animals respire through the lungs?
Type	multiple choice
Option	Earthworms and insects
Option	Sponges, coelenterates, and flatworms
Option	Fishes and aquatic arthropods
Option	Amphibians, reptiles, birds, and mammals
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Choose the incorrect pair from the following
Type	multiple choice
Option	Lizard (reptile) – lungs
Option	Salmon (Fish) – gills
Option	Whale (mammal) – gills
Option	Earthworm (annelid) – moist cuticle
Answer	3
Solution	
Positive Marks	4
Negative Marks	1

Question	Choose the correct sequence of respiration from the following: (I) Diffusion of O_2 and CO_2 between blood and tissues. (II) Diffusion of gases (O_2 and CO_2) across alveolar membrane (III) Breathing or pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out. (IV) Utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2 (V) Transport of gases by the blood.
Type	multiple choice
Option	$(I) \rightarrow (II) \rightarrow (III) \rightarrow (IV) \rightarrow (V)$
Option	$(V) \rightarrow (IV) \rightarrow (III) \rightarrow (II) \rightarrow (I)$
Option	$(IV) \rightarrow (I) \rightarrow (II) \rightarrow (V) \rightarrow (III)$
Option	$(III) \rightarrow (II) \rightarrow (V) \rightarrow (I) \rightarrow (IV)$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The structure which does not contribute to the breathing movement in mammals is
Type	multiple choice
Option	Larynx
Option	Ribs
Option	Diaphragm
Option	Inter-costal muscles
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Site of gaseous exchange in lungs are
Type	multiple choice
Option	Alveoli
Option	Tracheoles
Option	Bronchioles
Option	Pleura
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Match List I with List II & choose the correct option from the following: List-I / List-II A. Larynx / 1. Reduce friction on lung surface B. Alveoli / 2. Cartilaginous of lap C. Epiglottis / 3. Sound box D. Pleura / 4. Vascularised bag like structures
Type	multiple choice
Option	A-2 B-1 C-4 D-3
Option	A-3 B-2 C-4 D-1
Option	A-3 B-4 C-2 D-1
Option	A-3 B-2 C-1 D-4
Answer	3
Solution	
Positive Marks	4

Negative Marks	1
Question	The chamber formed dorsally by the vertebral column, ventrally by sternum, laterally by ribs and on the lower side by dome-shaped diaphragm is
Type	multiple choice
Option	Cranial chamber
Option	Pelvic chamber
Option	Abdominal Chamber
Option	Thoracic chamber
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	During inspiration, the volume of the thoracic cavity increases because of
Type	multiple choice
Option	Contraction of the diaphragm and external intercostal muscles
Option	Relaxation of the diaphragm and external intercostal muscles
Option	Contraction of diaphragm and relaxation of external intercostal muscles
Option	Relaxation of the diaphragm and contraction of external intercostal muscles
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which instrument helps in the clinical assessment of pulmonary functions?
Type	multiple choice
Option	Sphygmomanometer
Option	Stethoscope
Option	Spirometer
Option	Electrocardiograph
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	On an average healthy human breathes _____ times/minute.
Type	multiple choice
Option	12-16
Option	21-26
Option	30-32
Option	6-8
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following sequences is correct to initiate expiration? (I) Relaxation of external intercostal muscles and return of diaphragm and sternum to their normal position (II) Air expelled from lungs (III) Volume of thorax decreases (IV) Intrapulmonary pressure increases
Type	multiple choice
Option	I, III IV, II
Option	II, IV, III, I
Option	IV, III, II, I
Option	I, II, III, IV
Answer	1

Solution	
Positive Marks	4
Negative Marks	1
Question	Following illustration depicts the mechanism of breathing. In which of the following option all the parts A B, C, and D are correctly labeled? 
Type	multiple choice
Option	A-Air entering into lungs; B - Ribs and sternum raised; C- Diaphragm contracted; D-The volume of the thorax raised
Option	A- Air expelled from lungs; B – Ribs and sternum return to original position; CDiaphragm relaxed; D - Volume of thorax decreased
Option	A- Air expelled from lungs; B- Ribs and sternum raised; C - Diaphragm relaxed; D Volume of thorax decreased
Option	A - Air expelled from lungs; B - Ribs and sternum raised; C - Diaphragm contracted; D - Volume of thorax decreased
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Match List I with List II & Choose the correct option

List-I	List-II
(a) Expiration	1. Atmospheric air is drawn in
(b) Inspiration	2. Intrapulmonary is higher than atmospheric pressure
	3. Atmospheric air is drawn out
	4. Intra-pulmonary pressure is less than atmospheric pressure

Type	multiple choice
Option	a: 3,2 b: 1,4
Option	a: 3,4 b: 1,2
Option	a: 1,2 b: 3,4
Option	a: 1,4 b: 3,2
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Assertion: During inspiration, the volume of the thorax increases. Reason: This happens due to the relaxation of the diaphragm and inspiratory muscles.
Type	multiple choice
Option	If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
Option	If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
Option	If the assertion is true but the reason is false
Option	If both the assertion and reason are false
Answer	3
Solution	
Positive Marks	4

Negative Marks	1
Question	Which of the following sequences is correct to initiate inspiration? (I) The contraction of external intercostal muscles raises the ribs and sternum (II) Volume of the thorax increases in the dorsoventral axis (III) Intrapulmonary pressure decreases (IV) Diaphragm contraction (V) Air rushes into the lungs (VI) Volume of the thorax increases in the anterior-posterior axis
Type	multiple choice
Option	(i), (ii), (iv), (v), (iii), (vi)
Option	(i) (ii), ((iii),((iv), (vi), (v)
Option	(i) (ii), (iv), (vi), (iii), (v)
Option	(vi), (v), (i), (ii), (iii), (iv)
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	During expiration, the diaphragm becomes?
Type	multiple choice
Option	Dome-shaped (arched upwards)
Option	Oblique
Option	Concave
Option	Flattened
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following volume of air cannot be measured by a spirometer?
Type	multiple choice
Option	IRV
Option	ERV
Option	RV
Option	Both (a) and (b)
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Additional volume of air, a person can expire by a forcible expiration is called
Type	multiple choice
Option	TV
Option	ERV
Option	IRV
Option	EC
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Volume of air remaining in the lungs even after a forcible expiration is called
Type	multiple choice
Option	IRV
Option	RV
Option	FRC
Option	EC

Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Additional volume of air a person can inspire by a forcible inspiration is called
Type	multiple choice
Option	IC
Option	EC
Option	IRV
Option	ERV
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Total lung capacity is approximately
Type	multiple choice
Option	1200 ml
Option	2400 ml
Option	4000 ml
Option	5800 ml
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Residual volume is
Type	multiple choice
Option	Greater than vital capacity
Option	Greater than tidal volume
Option	Lesser than tidal volume
Option	Greater than inspiratory reserve volume
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is a part of diffusion membrane
Type	multiple choice
Option	Thick squamous epithelium of alveoli
Option	Endothelium of alveolar capillaries
Option	Basement membrane below endothelium
Option	All of the above
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statement is correct?
Type	multiple choice
Option	Alveoli are the primary sites of exchange of gases
Option	Exchange of gases occur at tissue level and alveolar surface.
Option	Pressure contributed by an individual gas in a mixture of gases is called partial pressure and is represented as pO_2 for oxygen and pCO_2 for carbon dioxide.
Option	All of the above
Answer	4
Solution	

Positive Marks	4
Negative Marks	1
Question	The amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher compared to that of O_2. Why?
Type	multiple choice
Option	Because diffusion of gases takes place only at alveoli
Option	Because the solubility of CO_2 is 20 – 25 times lesser than that of O_2
Option	Because the solubility of CO_2 is 20- 25 times higher than that of O_2
Option	Because alveoli are very large in number
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which condition is favourable for formation of oxyhaemoglobin?
Type	multiple choice
Option	High PO2, low PCO_2, Lesser H^+ concentration and lower temperature
Option	Low PO2, low PCO_2, Lesser H^+ concentration and lower temperature
Option	High PO2, high PCO_2, Lesser H^+ concentration and lower temperature
Option	Low PO_2, high PCO_2, Lesser H^+ concentration and lower temperature
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Select the incorrect statement – Statement I:- O_2 binds with haemoglobin in a irreversible manner to form oxyhaemoglobin Statement II:- Each haemoglobin molecule can carry maximum four molecules of O_2
Type	multiple choice
Option	Both statements are correct
Option	Both statements are incorrect
Option	Statement I is incorrect and statement II is correct
Option	Statement I is correct and II is incorrect
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	How much O2 is delivered to tissue by each 200 ml of oxygenated blood
Type	multiple choice
Option	5 ml
Option	10 ml
Option	15 ml
Option	20 ml
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	What is the value of pO_2 and pCO_2 in pulmonary vein?
Type	multiple choice
Option	pCO2 = 95 mmHg; pO_2 = 45 mmHg
Option	pO2 = 95 mmHg; pCO_2 = 40 mmHg
Option	pCO2 = 40 mmHg; pO_2= 45 mmHg

Option	pCO2 = 45 mmHg; pO_2= 45 mmHg
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Gaseous exchange takes place only at alveoli on respiratory surface because of-
Type	multiple choice
Option	Rich supply of capillaries at alveoli
Option	Large surface area of alveoli
Option	Very thin respiratory membrane
Option	All of the above
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following factors affect the diffusion of gases?
Type	multiple choice
Option	Partial pressure of diffusing gases
Option	Solubility of gases
Option	The thickness of diffusion membrane
Option	All of these
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statements about the partial pressure of CO_2 is true?
Type	multiple choice
Option	It is higher in alveoli than in pulmonary artery
Option	It is higher in the systemic arteries than in tissues
Option	It is higher in systemic veins than in systemic arteries
Option	It is higher in the pulmonary veins than in pulmonary arteries
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Arrange the following in the order of increasing volume (i) Tidalvolume (ii) Residual volume (iii) Expiratory reserve volume (iv) Vital capacity
Type	multiple choice
Option	(i) < (ii) < (iii) < (iv)
Option	(i) < (iii) < (ii) < (iv)
Option	(i) < (iv) < (iii) < (ii)
Option	(i) < (iv) < (ii) < (iii)
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Assertion: Vital capacity is higher in athletes than non - athletes. Reason: Vital capacity is about 3.5-4.5 litres in normal adult person.
Type	multiple choice
Option	If both the assertion and the reason are true and the reason is a correct explanations of the assertion

Option	If both the assertion and reason are true but the reason is not a correct explanation of the assertion
Option	If the assertion is true but the reasons is false
Option	If both the assertion and reason are false
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The total Lung Capacity (TLC) is the total volume of air accommodates in the lungs at the end of forced inspiration. This includes
Type	multiple choice
Option	RV (Residual Volume); IC (Inspiratory Capacity); EC (Expiratory Capacity); And ERV (Expiratory Reserve Volume)
Option	RV(Residual volume); ERV(Expiratory Reserve Volume); and EC (Expiratory Capacity)
Option	RV (Residual Volume); ERV (Expiratory Reserve Volume); VC (Vital Capacity) and FRC (Functional Residual Capacity)
Option	RV (Residual Volume); ERV(Expiratory Reserve Volume); TV (Tidal Volume); and IRV (Inspiratory Reserve Volume)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	How much air can be inspired or expired during normal breathing?
Type	multiple choice
Option	500 ml
Option	2500 ml
Option	1500 ml
Option	5500 ml
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following statement is incorrect?
Type	multiple choice
Option	A healthy man can inspire or expire approximately 6000 to 8000 ml of air per second.
Option	Total volume of air that can accommodated in the lungs at the end of forced inspiration is called total lung capacity.
Option	Total volume of air a person can expire after a normal inspiration is called expiratory capacity.
Option	Amount of air remain in lungs even after deep and forceful expiration is called residual volume.
Answer	1
Solution	
Positive Marks	4
Negative Marks	1